Exp No: 16 Experiment to understand EDA-Quantitative and Qualitative analysis.

AIM:

To understand EDA - Quantitative and Qualitative analysis.

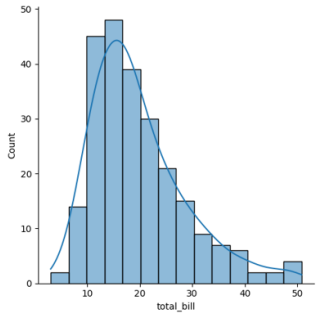
PROGRAM:

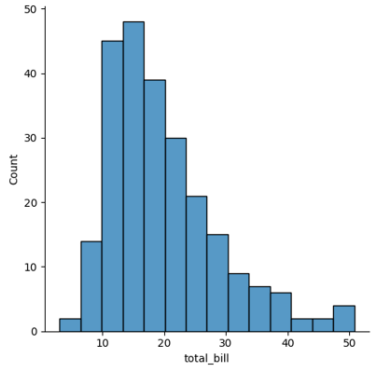
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
tips = sns.load_dataset('tips')
tips.head()
sns.displot(tips.total_bill, kde=True)
plt.show()
sns.displot(tips.total_bill, kde=False)
plt.show()
sns.jointplot(x=tips.tip, y=tips.total_bill)
plt.show()
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="reg")
plt.show()
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="hex")
plt.show()
sns.pairplot(tips)
plt.show()
```

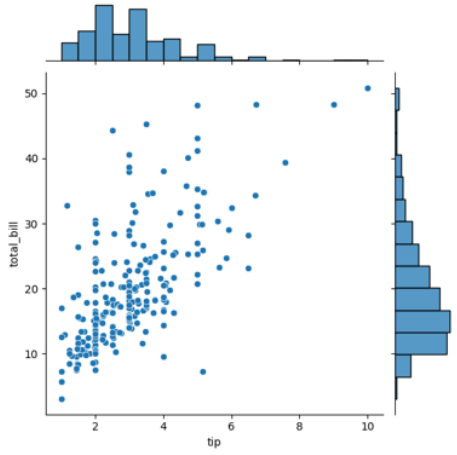
```
tips.time.value_counts()
sns.pairplot(tips, hue='time')
plt.show()
sns.pairplot(tips, hue='day')
plt.show()
sns.heatmap(tips.corr(numeric_only=True), annot=True)
plt.show()
sns.boxplot(y=tips.total_bill)
plt.show()
sns.boxplot(y=tips.tip)
plt.show()
sns.countplot(y=tips.day)
plt.show()
sns.countplot(y=tips.sex)
plt.show()
tips.sex.value_counts().plot(kind='pie')
plt.show()
tips.sex.value_counts().plot(kind='bar')
plt.show()
sns.countplot(y=tips[tips.time == 'Dinner']['day'])
plt.show()
```

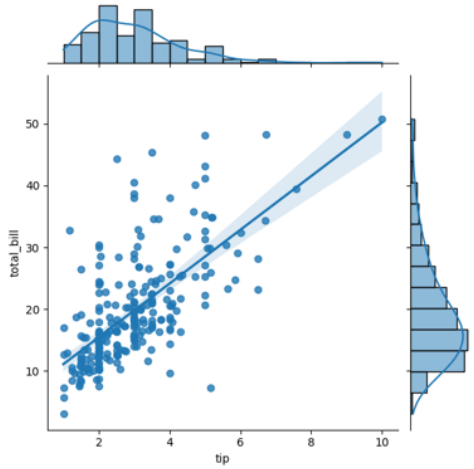
OUTPUT:

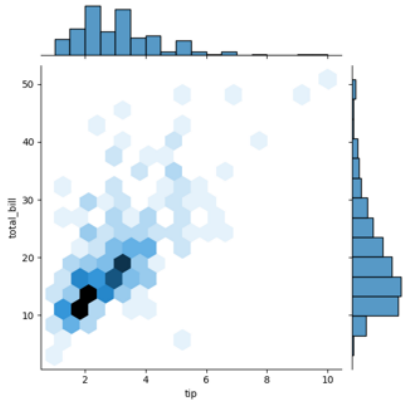
	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

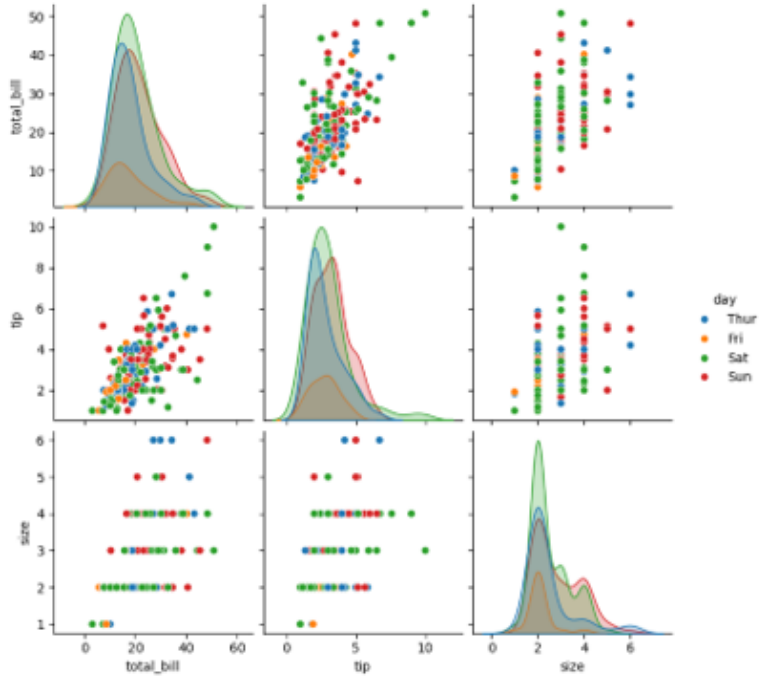


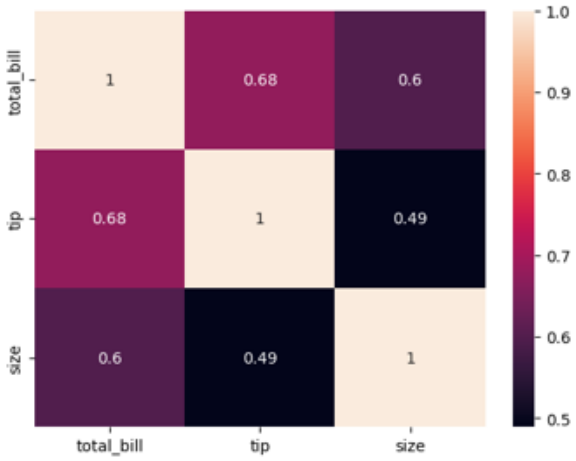


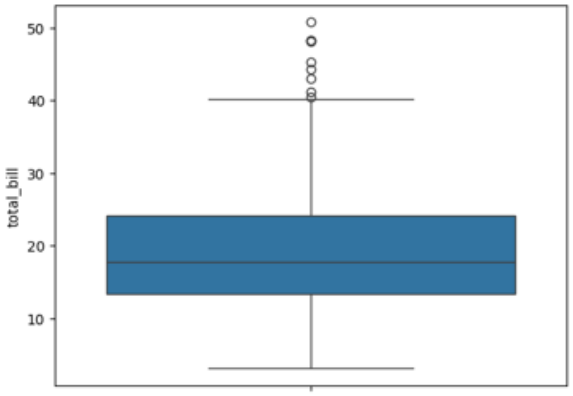


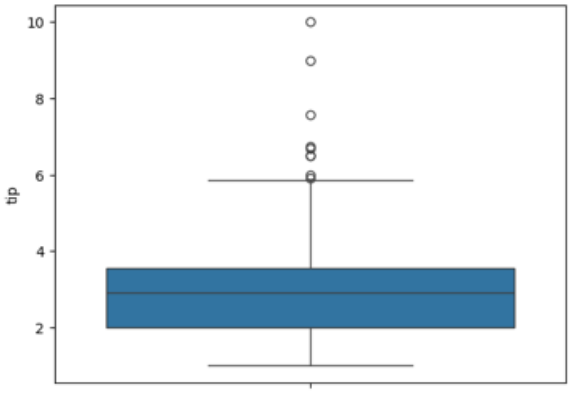

total_bill
tip

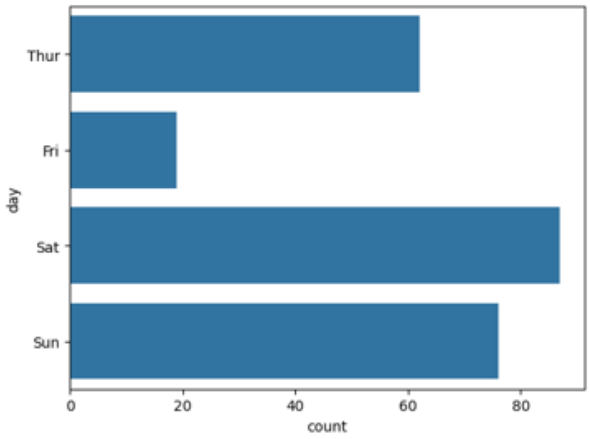

total_bill
tip

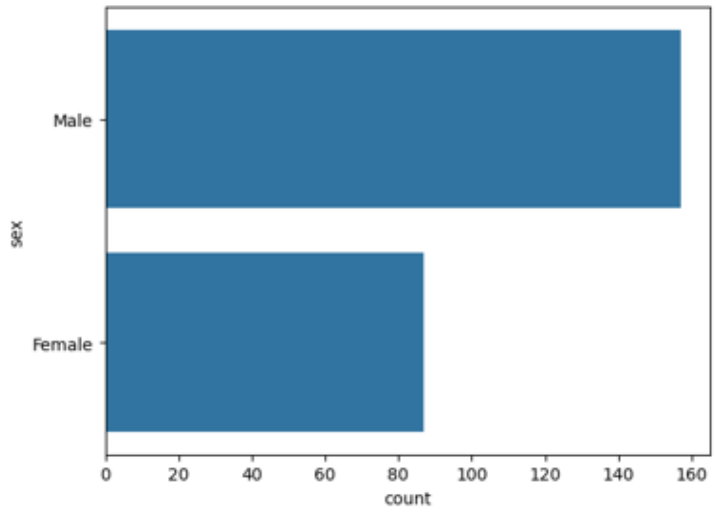

total_bill
tip
size
day
Thur
Fri
Sat
Sun

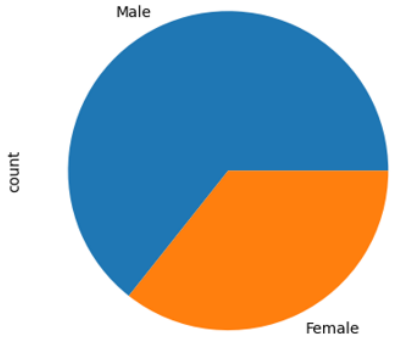

total_bill
tip
size
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
total_bill
tip
size
1.0
0.9
0.8
0.7
0.6
0.5

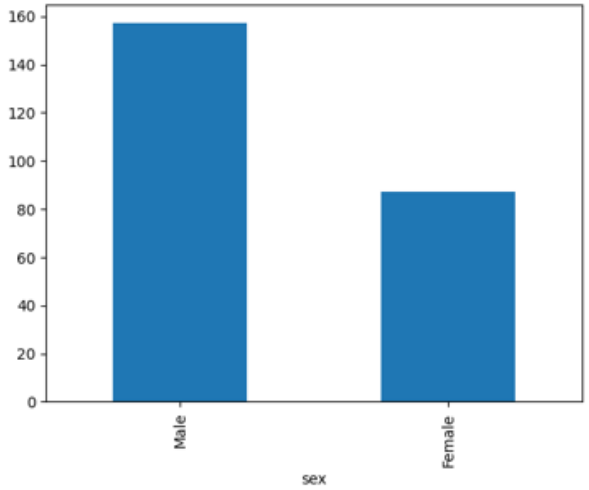

total_bill
50
40
30
20
10

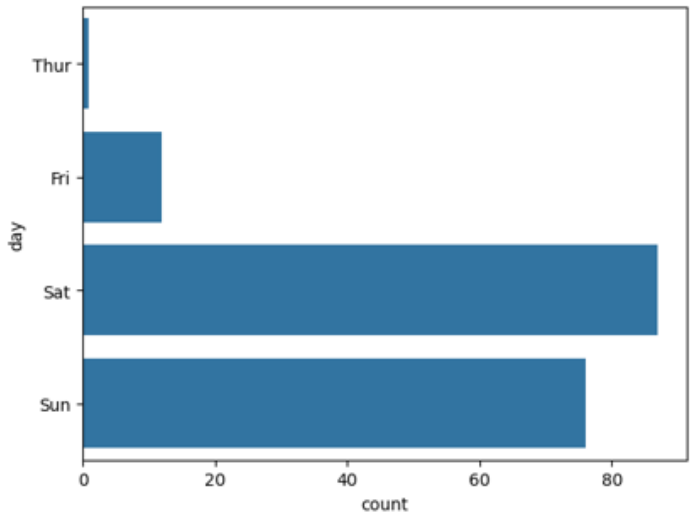

tip
10
8
6
4
2


day
Thur
Fri
Sat
Sun
0
20
40
60
80
count


Male
Female
sex
0
20
40
60
80
100
120
140
160
count


Male
Female
count


160
140
120
100
80
60
40
20
0
Male
Female
sex



RESULT:

Thus we have successfully understood EDA - Quantitative and Qualitative analysis.